MANASA RAGHAVARAJU

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PROFESSIONAL SUMMARY

Software Development Engineer with 2+ years of experience in web development, REST API design, and scalable system implementation. Strong foundation in Data Structures, Algorithms, and Object-Oriented Programming. Experienced in building cloud-based applications using AWS and developing reliable, high-performance software solutions in Agile environments.

EDUCATION

Master's in Computer Science (GPA – 4.00/4.00) ***University Of Nebraska – Lincoln (Jan 2024-Dec 2025)***

Bachelor of Technology in Computer Science (CGPA - 8.11/10.00) ***CMR Engineering College (2018- 2022)***

WORK EXPERIENCE

Software Engineer | University of Nebraska-Lincoln ***November 2025 – Present***

- Develop and optimize an ML-driven computer vision pipeline for agricultural automation, achieving **98% seed detection accuracy** through model tuning, dataset refinement, and training optimization using **YOLO, CNNs, OpenCV**, and labeled datasets prepared with CVAT.
- Support end-to-end system integration, including dataset preparation, model experimentation, hardware integration, real-time inference, and exporting results for analysis and reporting.
- Develop and maintain a university research website, implementing web features to support research data visualization and project communication.

Full Stack Developer | Faeiron Acrobatics ***May 2025 – August 2025***

- Developed and enhanced features for a React + AWS serverless web application using AWS Lambda, API Gateway, DynamoDB, and Cognito.
- Integrated secure signup with AWS Secrets Manager and designed role-based dashboard functionality for Employees and Guests.
- Built and integrated backend APIs for Inquiry management and Newsletter subscriptions, ensuring reliable SES email notifications.
- Collaborated on end-to-end deployment using **AWS CDK and CodePipeline**, troubleshooting integration, CORS, and UI issues.

Software Development Engineer I | LTIMindtree ***August 2022 – December 2023***

- As an Oracle Apps Developer, developed Oracle E-Business Suite applications, including customization, enhancements, and bug fixes.
- Developed and optimized **PL/SQL scripts** for various data manipulation, extraction, and validation, improving query performance by 30% and reducing data errors by 20%.
- Optimized **database queries** and supported deployment/testing of Oracle applications, reducing report generation time by 30% and improving reliability.

Intern | Nouveau Labs ***December 2021 – July 2022***

- Collaborated with the development team on a video conferencing application, gaining hands-on experience with **Elasticsearch** and contributing to the backend development.
- Worked on optimizing **search queries** and aggregation techniques to enhance search accuracy, reduce latency, and improve overall search performance for end-users.
- Fine-tuned **index structures** for faster data retrieval, cutting retrieval time from seconds to milliseconds on key queries.

SKILLS

Languages & Web:	C, C++, Java, Python, JavaScript, SQL, PL/SQL, TypeScript, HTML, CSS, Bootstrap, React JS, Angular JS
Backend & Architecture:	Spring Boot, Hibernate, REST APIs, Microservices, Serverless Computing
Databases:	MySQL, Oracle, DynamoDB
Cloud & DevOps:	AWS, CI/CD, Git, GitHub
Core Concepts:	Data Structures, Algorithms, OOP, DBMS, SDLC, Debugging, Unit Testing, Performance Optimization
Tools:	Jira, Linux, Windows
Professional Skills:	Problem Solving, Communication, Teamwork, Leadership, Organizational Skills

PROJECTS

Buyer-Seller Market for Books

- Tech Stack: Ethereum Blockchain, Solidity, Smart Contracts
- Worked as a Backend Developer in a team of four on a website model for an e-commerce application that provides a platform to buy, sell, and resell books by users on Blockchain to achieve decentralization and perform secure and efficient transactions.
- Used Ethereum, which facilitates secure and transparent transactions between buyers and sellers using smart contracts.

Seattle PD 911 Calls Analysis

- Tech Stack: Python, Pandas, Scikit-learn, XGBoost, Folium, GeoPandas, HDBSCAN.
- Analyzed over 1M emergency call records by integrating spatial, temporal, and weather data to uncover patterns in emergency response.
- Applied machine learning techniques such as clustering (HDBSCAN) and association rule mining, achieving 85% accuracy in identifying high-risk zones and peak emergency hours.

CERTIFICATIONS AND ACHIEVEMENTS

- AWS Certified Cloud Practitioner Certification | Amazon Web Services
- Ranked among the top teams in the Virtusa Codelite hackathon.